

Short Communication

Changes in Breast Cancer Incidence Rates in the United States by Histologic Subtype and Race/Ethnicity, 1995 to 2004

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Abstract

Breast cancer incidence rates rose throughout the 1980s and 1990s in the United States but have recently declined through 2004. Studies reporting this decline primarily attribute it to the sharp decline in menopausal hormone use following publication of the Women's Health Initiative trial results. However, they have not stratified rates by either histologic type or race/ethnicity, which could further inform contributors to these trends. Using data from 13 cancer registries that participate in the Surveillance, Epidemiology, and End Results program, we evaluated annual percent changes (APC) in breast cancer incidence rates from 1995 to 2004 by histologic type and race/ethnicity for intervals identified using joinpoint regression. Invasive ductal carcinoma and invasive lobular carcinoma incidence rates fell steadily from 1998 to 2004 [APC, -3.07% (95% confidence interval, -4.10 to -2.02) and APC, -3.18%

(95% confidence interval, -5.18 to -1.03), respectively]. Declines in rates of breast cancer overall and invasive ductal carcinoma were primarily limited to women ≥ 50 years of age and to non-Hispanic whites and Asian/Pacific Islanders, and declines in rates of invasive lobular carcinoma were primarily limited to non-Hispanic whites. The majority of these declines began around 1998 and all began before 2002 when the Women's Health Initiative trial results were published; thus, the abrupt decline in hormone therapy use starting in 2002 is unlikely to be primarily responsible for the recent decline in breast cancer rates. The declines observed thus far are likely attributable to saturation of screening, although further declines related to the widespread cessation of hormone use may follow. (Cancer Epidemiol Biomarkers Prev 2007;16(12):2773–80)

Introduction

We have previously reported that incidence rates of invasive breast cancer increased 4% from 1987 to 1999 in the United States and that there were substantial variations in incidence patterns by histologic type (1). Specifically, rates of invasive ductal carcinoma (IDC), the most common histologic type of breast cancer, increased 3% over this time period, whereas invasive lobular carcinoma (ILC) rates increased 52% and invasive ductal-lobular carcinoma (IDLC) rates increased 96%. Recent reports suggest that incidence rates of invasive breast cancer are declining in the United States (2–8). Specifically, based on Surveillance, Epidemiology, and End Results (SEER) data, an 8.6% decrease in incidence was observed from 2001 to 2004 (2).

One of the primary explanations offered for the declines observed is that they are due to the sharp 38% to 68% declines (3, 9, 10) in menopausal hormone use

that occurred after the publication of the Women's Health Initiative (WHI) estrogen and progestin trial results (11). Alternatively, Jemal et al. (5) argue that the decline is largely due to screening saturation given that the decline in rates began around 1999 across multiple age groups and were most pronounced for localized stage disease and tumors ≤ 2 cm in size. However, none of these studies reported trends stratified by factors such as histologic type and race/ethnicity, factors that have been found to be related to hormone use. It is noteworthy that although several studies have concluded that the primary explanation for the drop in breast cancer incidence rates is due to the widespread cessation of hormone therapy use that began after the WHI trial results were published in July 2002, the data reported in two of these studies indicate that rates have been dropping steadily since 2000 (6, 7), suggesting that factors other than changes in hormone use are responsible for these trends. Analyses by histologic type are particularly relevant given our previous observations about differences in breast cancer incidence trends by histologic type (1) and data that clearly and consistently suggest that combined estrogen and progestin hormone therapy (CHT) is associated with a 2.0- to 3.9-fold increased risk of lobular carcinomas (both ILC and IDLC) but not with IDC risk (12–17). The sharp rise in lobular carcinoma incidence rates through the 1990s is

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Table 1. Age-adjusted incidence rates of invasive breast carcinoma by histologic type, 1995 to 2004

Year	Total population in registries	All invasive cases		IDC	
		No. cases	Incidence rate per 100,000	No. cases	Incidence rate per 100,000
1995	10,707,977	23,161	223.2	15,975	154.4
1996	10,855,811	23,724	225.5	16,764	159.8
1997	11,018,678	24,888	231.7	17,316	161.6
1998	11,170,012	25,891	236.8	18,035	165.2
1999	11,315,376	26,246	235.8	18,057	162.5
2000	11,469,039	25,892	228.4	17,879	158.1
2001	11,665,882	26,581	229.8	17,555	152.0
2002	11,826,781	26,377	223.6	17,258	146.4
2003	11,970,333	24,998	207.8	16,584	138.1
2004	12,087,807	25,329	206.7	17,240	141.0
Number of joinpoints from 1995-2004*		1		1	
APC (95% CI)*		1995-1999	1.68% [†] (−0.38 to 3.78)	1995-1998	2.43% [†] (−0.68 to 5.64)
		1999-2004	−2.82% [†] (−4.26 to −1.36)	1998-2004	−3.07% [†] (−4.10 to −2.02)

Year	Total population in registries	All <i>in situ</i> cases		DCIS	
		No. cases	Incidence rate per 100,000	No. cases	Incidence rate per 100,000
1995	10,707,977	4,076	40.1	3,306	32.4
1996	10,855,811	4,268	41.4	3,450	33.4
1997	11,018,678	4,858	46	3,988	37.7
1998	11,170,012	5,630	52.2	4,609	42.7
1999	11,315,376	5,862	53.3	4,803	43.7
2000	11,469,039	5,936	52.9	4,865	43.4
2001	11,665,882	6,199	54.0	5,098	44.5
2002	11,826,781	6,314	53.9	5,147	43.9
2003	11,970,333	6,133	51.2	5,075	42.4
2004	12,087,807	6,339	51.9	5,352	43.9
Number of joinpoints from 1995-2004*		1		1	
APC (95% CI)*		1995-1999	8.31% [†] (4.92-11.80)	1995-1998	11.10% [†] (6.35-16.07)
		1999-2004	−0.94% [†] (−2.99 to 1.15)	1998-2004	0.38% [†] (−0.94 to 1.72)

*Joinpoints represent statistically significant changes in incidence rates based on joinpoint regression, and APCs are shown for the segments between joinpoints.

[†] APC is statistically different from zero at *P* < 0.05.

thought to be largely due to the strong association between CHT use and lobular carcinoma risk. The WHI estrogen and progestin trial did not find a difference in risk by histology, although the trial was underpowered to assess this relationship as it included only 38 ILC and 23 IDLC cases across both study arms.

The purpose of this study is to evaluate trends in breast cancer incidence rates by histologic type in the United States from 1995 to 2004, the past 10 years for which SEER data are available. If these declines are in fact largely attributable to the marked decrease in hormone therapy use, then one may expect that these declines would be largely confined to rates after 2002 and perhaps be more pronounced among lobular tumors.

Materials and Methods

Women (308,702) ≥30 years of age diagnosed with *in situ* or invasive breast cancer between 1995 and 2004 were identified through 13 population-based cancer registries in the United States that participate in the SEER Program. The underlying female population from which these cases were identified increased in size from 10,707,977 in 1995 to 12,087,807 in 2004. We included the registries serving Alaska Natives; Atlanta, Georgia; Connecticut; Detroit, Michigan; Hawaii; Iowa; Los Angeles, California;

New Mexico; Rural Georgia; San Francisco-Oakland, California; Seattle-Puget Sound, Washington; and Utah. These registries encompass ~14% of the U.S. population. This population is representative of the country as a whole with regard to socioeconomic status and education level, although it does include higher proportions of people living in urban areas and who are foreign born (18). Individual patient medical records are the source of data on patient and tumor characteristics. Informed consent was not obtained because cancer is a reportable disease by law, and release of information about all diagnoses made is mandatory, in the areas covered by SEER.

Estimates of the underlying populations covered by the SEER registries are based on data from the U.S. Bureau of the Census. Using SEER*Stat 6.3.5 (National Cancer Institute, Bethesda, MD; issued April 2007), these estimates were used to calculate incidence rates of breast cancer overall and by histologic type from 1995 to 2004 that were age adjusted to the 2000 U.S. population. Histologic classifications were based on the following International Classification of Diseases for Oncology Third Edition (ICD-O-3) codes: 8500 for IDC, 8520 and 8524 for ILC, and 8522 for IDLC, consistent with previous studies (19). Beginning in 2001, SEER switched from using the second edition of the ICD-O codes (ICD-O-2) to ICD-O-3. Codes 8500, 8520, and 8522 were used in both

Table 1. Age-adjusted incidence rates of invasive breast carcinoma by histologic type, 1995 to 2004 (Cont'd)

ILC		IDLC		Other	
No. cases	Incidence rate per 100,000	No. cases	Incidence rate per 100,000	No. cases	Incidence rate per 100,000
1,879	18.0	1,319	12.8	3,988	38.0
1,911	18.0	1,293	12.4	3,756	35.2
2,138	19.8	1,543	14.5	3,891	35.8
2,208	20.2	1,818	16.7	3,830	34.6
2,297	20.6	2,060	18.6	3,832	34.0
2,091	18.4	2,315	20.5	3,607	31.4
2,180	18.8	2,457	21.4	4,389	37.6
2,212	18.7	2,652	22.6	4,255	35.8
2,093	17.4	2,345	19.5	3,976	32.8
2,052	16.6	2,346	19.2	3,691	30.0
1		1		0	
1995-1998	5.07% (−1.44 to 12.02)	1995-2001	10.85% [†] (7.54-14.25)	1995-2004	−1.41% (−3.00 to 0.21)
1998-2004	−3.18% [†] (−5.28 to −1.03)	2001-2004	−5.05% (−12.18 to 2.65)		

ICD-O editions and defined identically in both versions. Code 8524 was new to ICD-O-3 and is defined as “infiltrating lobular mixed with other types of carcinoma,” but it represents a subset of tumors that would have been coded as 8520 using ICD-O-2. However, this code is uncommonly used as it accounted for only 520 (6%) of the 8,537 ILC cases included in this study that were diagnosed since 2001. Among *in situ* carcinoma cases, those with the following ICD-O-3 codes were classified as ductal carcinoma *in situ* (DCIS) consistent with previous studies: (20) 8201, 8230, 8500, 8501, 8503, 8507, and 8523. Race/ethnicity was categorized into five groups: non-Hispanic whites, blacks, American Indians/Alaska Natives, Asians/Pacific Islanders, and Hispanic whites.

Statistically significant changes in incidence rates over this time period were evaluated using joinpoint regression (21) with the Joinpoint regression software developed by the Statistical Research and Applications Branch at the National Cancer Institute. This approach uses a weighted least squares log-linear approach to identify time points when changes in incidence rates occur. In evaluating incidence rate trends by histologic type, age, and race/ethnicity, we allowed up to two joinpoints in each regression model. The best-fitting model was chosen based on permutation tests that maintain an overall two-sided *P* value of <0.05. Annual percent changes (APC) were calculated for each joined line segment starting in 1995 and then between any identified joinpoints through 2004. Heteroscedastic errors were assessed based on the number of cases in each year and strata using a Poisson model and used to calculate 95% confidence intervals (95% CI) for each APC.

Results

From 1995 to 1999, incidence rates of invasive breast cancer increased 1.68% (95% CI, −0.38 to 3.78) per year and, from 1999 to 2004, declined 2.82% (95% CI, −4.26 to −1.36) per year (Table 1). Similarly, IDC and ILC rates increased from 1995 to 1998 [2.43%/year (95% CI, −0.68 to 5.64) and 5.07%/year (95% CI, −1.44 to 12.02), respectively] and declined from 1998 to 2004 [3.07%/

year (95% CI, −4.10 to −2.02) and 3.18%/year (95% CI, −5.28 to −1.03), respectively]. IDLC rates increased sharply, 10.85%/year (95% CI, 7.54-14.25), from 1995 to 2001, but then declined somewhat from 2001 to 2004 [−5.05%/year (95% CI, −12.18 to 2.65)]. Incidence rates of invasive tumors with other or unknown histologic types held essentially constant over this time period (APC, −1.41%; 95% CI, −3.00 to 0.21). In contrast to the rates of invasive tumors, rates of *in situ* carcinomas and DCIS increased considerably from 1995 to 1999 and 1995 to 1998, respectively [all *in situ* cases: APC, 8.31% (95% CI, 4.92-11.80) and DCIS: APC, 11.10% (95% CI, 6.35-16.07)], but remained essentially constant afterwards through 2004 [all *in situ* cases: APC, −0.94% (95% CI, −2.99 to 1.15) and DCIS: APC, 0.38% (95% CI, −0.94 to 1.72)]. All of these trends were essentially the same and occurred over the same time intervals for women 50 to 79 years of age, which is an age group of women with high prevalences of menopausal hormone use and annual mammographic screening (Fig. 1).

There was some variation in these trends by age and race/ethnicity. Rates of invasive breast cancer overall, IDC, and ILC were essentially constant from 1995 to 2004 among women ≤49 years of age (Table 2). From 1998 to 2004, rates of invasive breast cancer overall declined 3.84%/year (95% CI, −5.62 to −2.03) among women 50 to 59 years of age and 2.77%/year (95% CI, −4.26 to −1.26) among women 70 to 79 years of age. A 5.60%/year (95% CI, −10.30 to −0.65) decline was observed among women 60 to 69 years of age from only 2001 to 2004. When assessed by histologic type, IDC rates fell 4.48%/year (95% CI, −6.41 to −2.51) from 1999 to 2004 for 50- to 59-year-old women, 4.62%/year (95% CI, −8.12 to −0.99) from 2000 to 2004 for 60- to 69-year-old women, and 3.98%/year (95% CI, −5.09 to −2.86) from 1998 to 2004 for 70- to 79-year-old women. In contrast, ILC rates remained essentially constant from 1995 to 2004 across each age strata. IDLC rates increased anywhere from 5.93%/year to 14.14%/year either across the entire study period or through 2000, 2001, or 2002 across all age groups. Recent declines in IDLC rates were observed for women 50 to 79 years of age, but none of these declines were statistically significant. Rates of *in situ* carcinomas and DCIS among women 40 to 79 years of age essentially mirrored the overall trends across women of all ages, as

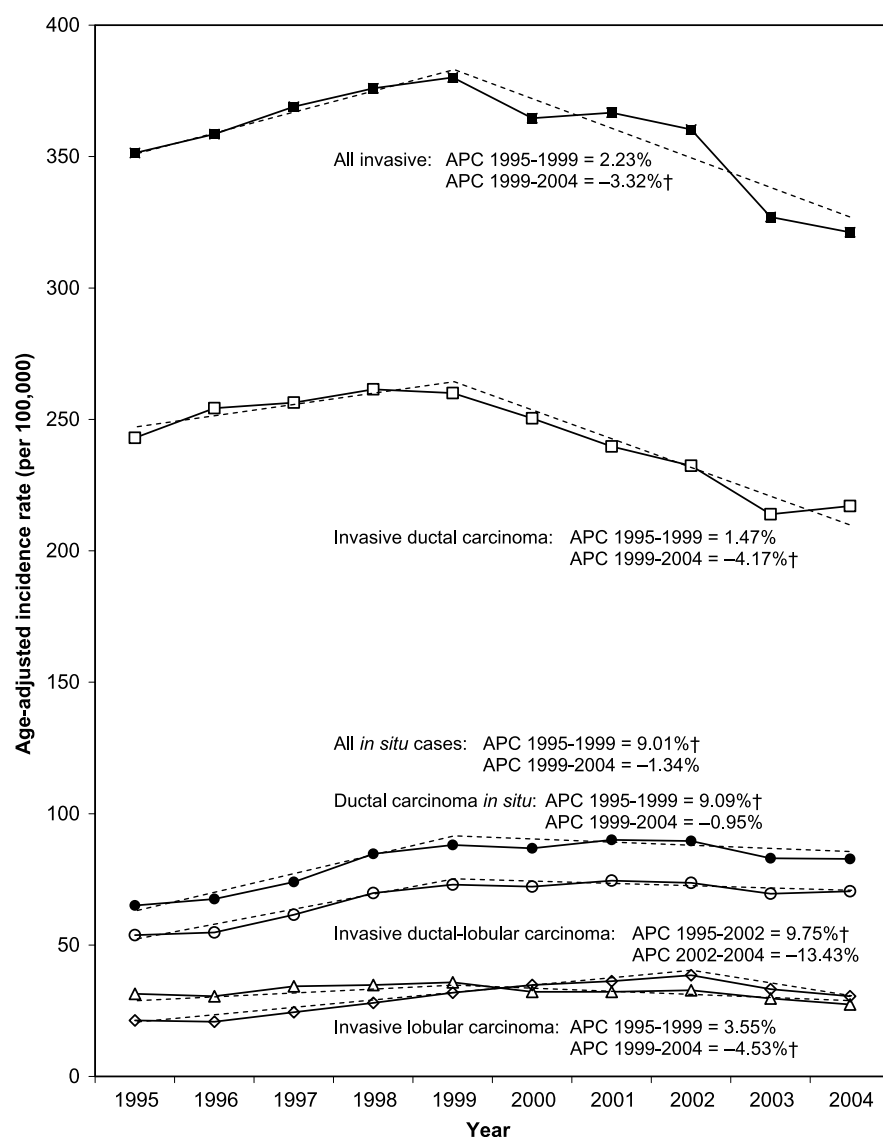
rates of *in situ* carcinoma rose from 1995 through about 1999 and have since held relatively constant.

With respect to race/ethnicity, rates of invasive breast cancer overall and ILC rates experienced statistically significant declines only among non-Hispanic whites, with a 2.89%/year (95% CI, -4.60 to -1.14) decrease observed from 1999 to 2004 for breast cancer overall and a 3.16%/year (95% CI, -5.47 to -0.79) decrease observed from 1998 to 2004 for ILC. Rates of IDC decreased only among non-Hispanic whites (APC, -3.20; 95% CI, -4.28 to -2.12) and Asian/Pacific Islanders (APC, -1.45; 95% CI, -2.85 to -0.03) from 1998 to 2004 and 1997 to 2004, respectively. IDLC rates climbed through 2001 for non-Hispanic whites and through 1999 for blacks (after which

nonstatistically significant declines in rates were observed) but have continued to rise for both Asian/Pacific Islanders and Hispanic whites through 2004. Rates of *in situ* carcinoma and DCIS increased through 1999 for non-Hispanic whites and through 1998 for Hispanic whites and have since stabilized, whereas they rose continuously from 1995 to 2004 for blacks and Asians/Pacific Islanders.

Discussion

After rising steadily from the 1980s through 1998 (1), incidence rates of invasive breast cancer declined 8% from 2000 to 2004. Authors of prior studies (2, 3, 6-8) also



* Black square = All invasive cases; white square = invasive ductal carcinoma; white triangle = invasive lobular carcinoma; white diamond = invasive ductal-lobular carcinoma; black circle = all *in situ* cases; and white circle = ductal carcinoma *in situ*. Dashed lines represent straight-line segments between joinpoints identified using joinpoint regression.

† Annual Percent Change (APC) is statistically significantly different from zero at $p < 0.05$.

Figure 1. Age-adjusted breast cancer incidence rates for women 50 to 79 years of age, 1995 to 2004. ■, all invasive cases; □, IDC; △, ILC; ◇, IDLC; ●, all *in situ* cases; ○, DCIS. Dashed lines, straight-line segments between joinpoints identified using joinpoint regression. †, APC is statistically significantly different from zero at $P < 0.05$.

documenting the recent decline in these rates have hypothesized that it may largely be due to the sharp 38% to 68% declines (3, 9, 10) in the use of estrogen and progestin therapy that occurred after the results of the WHI trial showing the harms associated with estrogen and progestin use were published in 2002 (11). The data presented here do not strongly support this explanation and instead support the hypothesis put forth by Jemal et al. (5) that saturation of screening may be primarily responsible for these declines.

Most importantly, the decline in rates, both overall and by histologic type, clearly began before the results of the WHI trial were published in 2002. This has also been documented in two recent studies assessing these trends, which found that rates began to decline in 2000 (6, 7). Here, we found that both IDC (the most common histologic type of breast cancer) and ILC rates have dropped steadily and at a fairly consistent rate since peaking in 1998. Thus, factors other than the widespread cessation of hormone therapy that began during the last half of 2002 must be at least as important and likely more important contributors to these trends.

What is more likely accounting for the decline in rates is saturation of screening. As argued by Jemal et al. (5), the observations that these declines began around 1999, occurred across all women ≥ 50 years of age, and were more pronounced for localized disease and tumors ≤ 2.0 cm all support this hypothesis. The data presented here further support this point given that declines in rates of both IDC and ILC were observed beginning in 1998. In addition, as shown here and previously (5, 20), whereas invasive rates have fallen, *in situ* carcinoma rates have held constant. Mammography is effective in identifying *in situ* tumors, and now that mammography rates have stabilized, and in fact started to fall, it may be that rates of invasive cancer are falling because *in situ* rates have stabilized. In other words, mammography is having its intended effect as *in situ* lesions are being detected by mammography before they develop into invasive tumors. The data support this as rates of invasive cancer began to fall when *in situ* rates began to stabilize.

There are also several observations reported here that are difficult to reconcile with the hormone therapy hypothesis. Although clear data on differences in the latency of the effects of hormone therapy on the development of ILC versus IDC are not available, given the strong evidence that estrogen and progestin therapy is more strongly related to risk of lobular carcinomas than it is to risk of IDC (12–17), one might expect rates of ILC to decline more rapidly than those of IDC after 2002. However, this was not observed, further suggesting that the widespread abrupt cessation in CHT use may not account for the trends observed. Much attention has been paid to the observation that this recent decline is only observed among hormone receptor-positive but not hormone receptor-negative disease (2, 5). However, it is difficult to interpret these data given the concurrent steady reductions of $\sim 50\%$ in the proportion of tumors with an unknown ER status within the populations studied. One study attempted to address this problem using multiple imputation (2), but there are inherent biases in using this strategy (22), particularly when the proportion of the variable with missing data is changing so rapidly over the study

period, and the methodologic approach used was not described.

A second reason why termination of CHT use is unlikely to account for the observed trends is the questionable biological plausibility of such an association. Based on studies evaluating the relationship between time since last use of menopausal hormone therapy and breast cancer risk, it is unlikely that a decline in incidence rates would be observed so soon after large numbers of women stopped using these hormones. The report by the Collaborative Group on Hormonal Factors in Breast Cancer evaluated risk among former users who had stopped using hormones for 5 years or longer and found that these women do not have an increased risk of breast cancer (23). However, few studies have evaluated shorter intervals since cessation of use. This has been evaluated closely in the Nurses' Health Study, which observed that former users of hormone therapy for 5 years or longer maintained a 44% elevation in their risk of breast cancer over the 2 years after they stopped using hormones (24). This is very similar to the 46% increased risk of breast cancer experienced by current users of hormone therapy in this study and indicates that breast cancer risk remains elevated among hormone users at least 2 years after their use ceases. In addition, the Million Women's Study observed that former users of hormone therapy who had stopped using hormones within 5 years maintained a 1.15-fold (95% CI, 1.07–1.24) elevated risk of breast cancer (although this analysis was not stratified by either type of hormone therapy or histologic type of breast cancer; ref. 25).

Third, we found that breast cancer rates only declined among non-Hispanic white women but not among women of other races/ethnicities. Wei et al. (10) evaluated changes in the prevalence of estrogen and progestin use by race and observed that, after the WHI trial results were published, the prevalence of use fell equally across all races, declining 44.1% among whites, 41.6% among African-Americans, and 38.4% among Asians. Prevalence of use is known to vary by race, but in this study, baseline rates were relatively similar as 14.3% of whites were current CHT users before the publication of the WHI trial results compared with 10.1% of African-Americans and 12.5% of Asians. Despite these differences, the complete lack of a decline in rates of breast cancer among African-Americans, Asians/Pacific Islanders, and Hispanic whites observed here argues against cessation of hormone therapy being primarily responsible for this trend.

Breast cancer incidence rates have declined in the United States since 1999 for two of the most common histologic types of breast cancer, IDC and ILC, which accounted for 77% of the invasive tumors included in this study. This decline predates the publication of the WHI trial results in 2002, and thus, these data do not support the hypothesis that the marked decline in rates of hormone therapy use that followed is a major contributor to this trend. These current trends are most likely attributable to saturation of mammography, although, given the nature of the descriptive data used here, causation can certainly not be inferred. A major limitation of this study is that it is descriptive and does not include any individual data on use of hormone therapy or use of mammographic screening. Nevertheless, although the decline in hormone therapy use may

Table 2. Trends in the incidence rates of IDC, ILC, and IDLC by age and race/ethnicity

Characteristic	All invasive		IDC	
	No. joinpoints	APC for time interval	No. joinpoints	APC for time interval
Age				
30-39	0	1995-2004: 0.70%	0	1995-2004: 0.65%
40-49	0	1995-2004: -0.57%	1	1995-1998: 1.20%
				1998-2004: -1.81%
50-59	1	1995-1998: 2.17%	1	1995-1999: 1.61%
		1998-2004: -3.84%*		1999-2004: -4.48%*
60-69	1	1995-2001: 1.73%*	1	1995-2000: 1.34%
		2001-2004: -5.60%*		2000-2004: -4.62%*
70-79	1	1995-1998: 2.57%	1	1995-1998: 2.22%
		1998-2004: -2.77%*		1998-2004: -3.98%*
80+	0	1995-2004: -1.83%*	1	1995-1998: 2.22%
				1998-2004: -4.15%
Race/ethnicity				
Non-Hispanic	1	1995-1999: 2.21%	1	1995-1998: 2.85%
White		1999-2004: -2.89%*		1998-2004: -3.20%*
Black	0	1995-2004: -0.50%	2	1995-1999: 1.22%
				1999-2001: -6.37%
				2001-2004: 2.87%
American Indian/Alaska Native	0	1995-2004: -2.40%	0	1995-2004: -2.09%
Asian/Pacific Islander	0	1995-2004: 0.16%	1	1995-1997: 8.28%
				1997-2004: -1.45%*
Hispanic white	0	1995-2004: -0.73%	0	1995-2004: -1.01%
Characteristic	All <i>in situ</i>			
	No. joinpoints	APC for each time interval		
Age				
30-39	2			1995-1998: 8.90%
				1998-2003: -4.49%
				2003-2004: 16.17%
40-49	2			1995-1998: 9.87%*
				1998-2001: -1.24%
				2001-2004: 2.49%
50-59	1			1995-1999: 7.89%*
				1999-2004: -1.89%
60-69	1			1995-2000: 8.56%*
				2000-2004: -2.13%
70-79	1			1995-1999: 9.09%*
				1999-2004: -1.39%
80+	0			1995-2004: 4.35%*
Race/ethnicity				
Non-Hispanic	1			1995-1999: 8.80%*
White				1999-2004: -1.12%
Black	0			1995-2004: 2.80%*
American Indian/Alaska Native	0			1995-2004: 2.60%
Asian/Pacific Islander	0			1995-2004: 4.76%*
Hispanic white	1			1995-1998: 12.47%*
				1998-2004: 0.55%

*APC is statistically different from zero at $P < 0.05$.

not be a major contributor to these trends thus far, the true effect of this decline on breast cancer rates may yet to be observed based on the risks of breast cancer former hormone user experience by time since they last used hormones. Given that breast cancer risk likely remains elevated for at least 2 years since stopping hormone use (24), but returns to baseline by 5 years since stopping (23), it is reasonable to speculate that breast cancer rates may start to be affected by declining hormone use starting in 2005 and may continue to be affected through 2008. Thus, continued close monitoring of these rates is of public health importance.

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Table 2. Trends in the incidence rates of IDC, ILC, and IDLC by age and race/ethnicity (Cont'd)

ILC		IDLC	
No. joinpoints	APC for time interval	No. joinpoints	APC for time interval
0	1995-2004: -2.18%	0	1995-2004: 6.14%*
0	1995-2004: 0.11%	0	1995-2004: 5.93%*
0	1995-2004: -0.67%	1	1995-2000: 12.32%*
0	1995-2004: -0.84%	1	2000-2004: -3.39%
0	1995-2004: -1.84%	1	1995-2001: 14.14%*
0	1995-2004: -0.84%	1	2001-2004: -7.80%
1	1995-1998: 6.66%	1	1995-2002: 8.76%*
0	1998-2004: -3.16%*	0	2002-2004: -12.78%
0	1995-2004: -0.79%	0	1995-2004: 6.32%*
0	1995-2004: 2.81%	0	1998-2004: -4.15%
0	1995-2004: 1.51%	1	1995-2001: 11.31%*
0	1995-2004: -1.87%	1	2001-2004: -5.20%
		0	1995-1999: 16.78%*
		0	1999-2004: -2.51%
		0	1995-2004: 1.89%
		1	1995-1997: 35.11%
		0	1997-2004: 6.69%*
		0	1995-2004: 6.16%*
DCIS			
No. joinpoints	APC for each time interval		
0	1995-2004: 0.76%		
2	1995-1998: 11.96%*		
	1998-2002: -1.13%		
	2002-2004: 5.45%*		
1	1995-1999: 8.20%*		
	1999-2004: -1.65%		
1	1995-2000: 8.17%*		
	2000-2004: -1.38%		
1	1995-1999: 9.76%*		
	1999-2004: -1.15%		
0	1995-2004: 4.79%*		
1	1995-1999: 9.00%*		
	1999-2004: -0.88%		
0	1995-2004: 3.58%*		
0	1995-2004: 0.95%		
0	1995-2004: 4.92%*		
1	1995-1998: 12.98%*		
	1998-2004: 1.70%		

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